

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458854.9074 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.043
Transit depth (Rp/Rs)^2	1.11 +/- 0.91 [%]
Orbital Inclination (inc)	80.68 +/- 2.91
Airmass coefficient 1 (a1)	0.19 +/- 0.11
Airmass coefficient 2 (a2)	1.76 +/- 0.47
Scatter in the residuals of the lightcurve fit is	51.26 %
Best Comparison Star	None
Optimal Aperture	2.3
Optimal Annulus	14.57
Transit Duration (day)	0.032 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.105 ± 0.043 vs 0.1214 (deviation 0.2σ)
Duration fit vs published	0.032 d vs 0.0742 d (deviation 0.67σ)
Mid-time O–C	6.5 min
Depth SNR	1.3
Residual scatter	51.26 %

Light curve

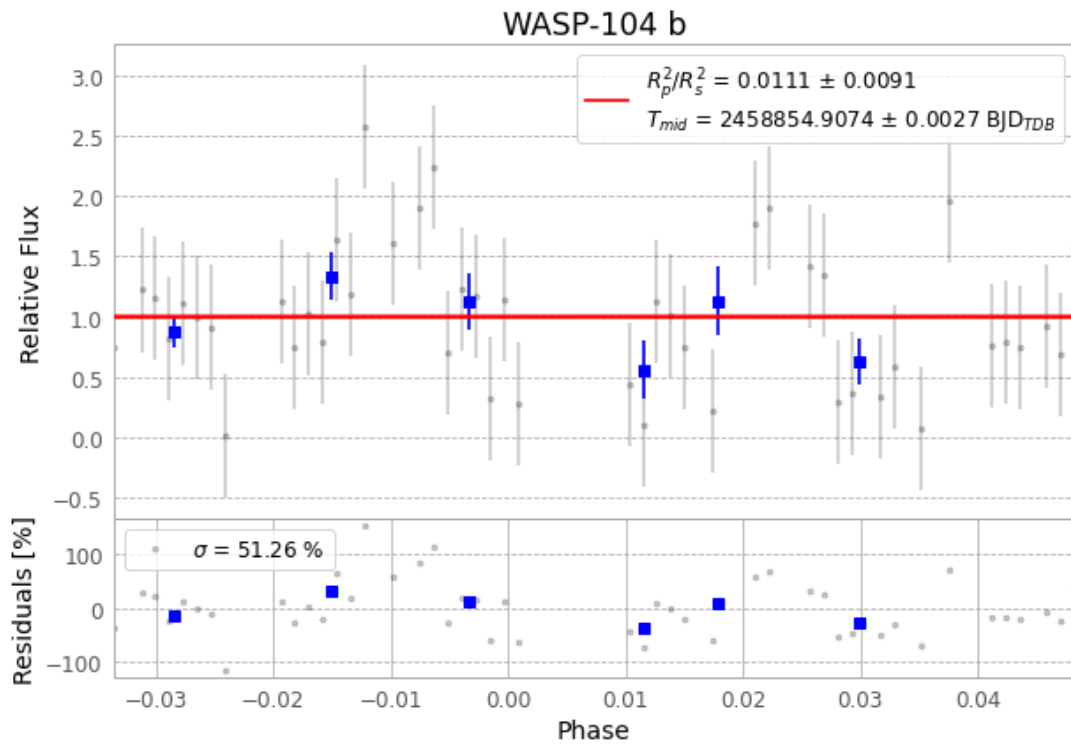


Figure 1 — FinalLightCurve_WASP-104 b_2020-01-06.png (SHA-256 ef4cc32960455140...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 334], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d1a2e594f44366b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), WASP-104200106081512.FITS → WASP-104200106114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-200106.log (a27cfbf9ed71...), FinalLightCurve_WASP-104 b_2020-01-06.png (ef4cc3296045...), FinalLightCurve_WASP-104 b_2020-01-06.csv (e64aec53b188...)

Compute resources

Wall time 138.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 01:18Z.